

Project Development Phase

Data Preprocessing and Business Questions

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Project Name

Housing Data Analysis using Tableau

1. Introduction

Data preprocessing is one of the most important steps in any data analytics project. Raw data often contains missing values, duplicate records, inconsistent formats, and unnecessary attributes. Before creating dashboards in Tableau, the housing dataset was cleaned and organized to ensure accurate analysis and reliable visualizations.

The cleaned dataset was then imported into Tableau Desktop to create interactive dashboards and business insights.

2. Dataset Description

The project uses a Housing Sales Dataset containing information about residential properties.

The dataset includes various attributes such as:

- Sale Price
- Bedrooms
- Bathrooms
- Floors
- House Age
- Basement Area
- Renovation Status
- Living Area
- Lot Area
- Latitude

- Longitude
- Year Built
- Year Renovated

These attributes help analyze factors influencing housing prices.

3. Data Preprocessing

The following preprocessing activities were performed before visualization:

- **Data Cleaning**

- Checked for missing values.
- Removed unnecessary or duplicate records.
- Corrected inconsistent data formats.

- **Data Validation**

- Verified numerical fields.
- Checked for invalid or unrealistic values.
- Ensured data consistency.

- **Data Formatting**

- Converted data into appropriate data types.
- Formatted date and numeric fields.
- Verified categorical variables.

- **Data Import**

- Imported the cleaned CSV dataset into Tableau Desktop.
 - Verified field names and data types.
 - Created calculated fields where required.
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4. Business Questions

The project aims to answer the following business questions:

1. What is the average selling price of houses?
2. Does house renovation increase selling price?
3. How does basement area affect housing prices?
4. What is the relationship between house age and selling price?
5. Which property characteristics contribute most to higher prices?

6. How are housing prices distributed across different properties?
 7. Which factors influence the housing market the most?
 8. How can interactive dashboards help users make better decisions?
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5. Development Process

The project development was carried out in the following stages:

1. Dataset Collection
 2. Data Cleaning
 3. Data Preprocessing
 4. Importing Data into Tableau
 5. Creating KPIs
 6. Developing Charts
 7. Building Interactive Dashboards
 8. Creating Story Points
 9. Testing Dashboard Functionality
 10. Final Dashboard Preparation
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6. Expected Outcome

The developed Tableau dashboard provides users with interactive visualizations for analyzing housing market trends. It enables quick identification of important patterns, supports data-driven decision-making, and simplifies the interpretation of complex housing data through KPIs, charts, filters, and dashboards.